



UNIVERSITY OF LAY ADVENTISTS OF KIGALI

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Faculty of Computing and Information Sciences

Course Code & Name: EWA408510 – E-Commerce and Web Application | Final Examination (Project-Based)
Academic year **2025-2026**

Instructors: Eric Maniraguha
Group: Session Day & Evening
Period: 1 June - 08 June, 2026

Exam Duration: 7 Days
Max Marks: 30 Marks

Instructions:

- *Attempt all tasks.*
 - *You may start with any section but clearly indicate what you are answering.*
 - *Read all requirements carefully before starting.*
 - *Present your work clearly and professionally.*
 - *Do not use electronic devices during assessment discussions (phones, smart watches, etc.).*
 - *Ensure all required information is correctly submitted.*
 - *Plagiarism or inability to explain your work will result in a score of zero.*
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Project Scenario

A local business in Rwanda wants to expand its operations by selling products and services online. The business has approached you as a web developer to design and implement a modern e-commerce platform that enables customers to browse products, add items to a shopping cart, place orders, and manage purchases online.

Your task is to design, develop, deploy, and present a complete **E-Commerce Web Application** that addresses the needs of an online business. You may choose any type of business, such as a bookstore, fashion store, electronics shop, pharmacy, restaurant ordering platform, grocery store, handicraft marketplace, or another approved e-commerce concept.

Project Requirements

1. User Interface (UI)

- Responsive and professional design
- Homepage with navigation menu

- Mobile-friendly interface
- 2. Product Management**
 - Product listing
 - Product details page
 - Product categories
- 3. Shopping Cart**
 - Add/remove items
 - Update quantities
 - Calculate totals
- 4. Checkout Process**
 - Customer details
 - Order summary
 - Order confirmation
- 5. Database Integration**
 - Products storage
 - Customer data
 - Orders management
- 6. GitHub (Mandatory)**
 - Project must be hosted on GitHub
 - Meaningful commit history required
- 7. Deployment (Mandatory)**
 - Application must be deployed online
 - Must be accessible during evaluation
- 8. CI/CD (Mandatory)**
 - Implement CI/CD pipeline (GitHub Actions or equivalent)
 - Automatic build/test/deploy on push
- 9. Docker (Mandatory)**
 - Application must be containerized using Docker
 - Dockerfile required
 - docker-compose required if multiple services exist
 - Evidence of build and execution must be provided

Deliverables

Each student must submit:

1. GitHub Repository Link
2. Live Deployment URL
3. Project Report (3–5 Pages)
4. Source Code
5. Database Script/File (if applicable)

6. Screenshots of the Application

Project Report Structure

Your report should include:

- Introduction
- Problem Statement
- Objectives
- System Features
- Technologies Used
- System Architecture
- Screenshots
- GitHub Repository Link
- Deployment Link
- CI/CD Description
- Challenges Encountered
- Future work
- Conclusion

Innovation Bonus (Up to 5 Marks)

Students may earn additional marks for implementing innovative features such as:

- Payment Gateway Integration (e.g., Mobile Money, Stripe, PayPal)
- AI-powered recommendations
- Analytics Dashboard
- Advanced Security Features
- Multi-vendor Marketplace Features
- Unique and Creative Business Ideas

Important Notes

- The project must be a Web Application. Desktop applications are not permitted.
- Students may use any programming language, framework, database, or hosting platform.
- Students will be required to present and demonstrate their application.
- During the oral defense, students will be asked questions about their design decisions, code, database structure, deployment process, and CI/CD implementation.
- Students must be able to explain all aspects of their project.
- Evidence of plagiarism or inability to explain the submitted work may result in a score of zero.
- The project must be completed and submitted within seven (7) days.

Closing Remark

As you begin this final project, I encourage you to approach your work with honesty, discipline, and commitment.

*The Bible reminds us in **Colossians 3:23**:*

“Whatever you do, work at it with all your heart, as working for the Lord, not for human masters.” This means that whatever task you are given should be done with full effort and sincerity, as if you are doing it before God. It teaches us to value integrity, hard work, and excellence in everything we do, not just for grades or recognition, but as a reflection of good character.

I wish you wisdom and focus as you complete your project.

environments. I encourage you to continue applying these skills as you progress in building and analyzing e-commerce platforms and web-based applications.

I wish you continued success in your academic journey and in all your future learning and practice in the field of digital business and web technologies.

Keep striving for excellence in all you do.